

Serpentine Laser Array

Polarization-Controlled Discrete Mode Switching (1x1 to 6x8 Expansion Study)

Red laser dot(s) projected onto a diffuse white LED panel surface across 20 trials in 4 sessions. **Key finding (Session 4):** orthogonal polarizer film at the input port produces discrete satellite modes even in a 1x1 array — confirming that discrete mode generation is driven by **polarization state**, not array size. Expansion to **6x8** remains planned for large-scale collective propagation.

Session 1 — White LED Panel, No Polarizer (videos 01–06)

File	Dur.	Description
01_baseline_single_dot.mp4	~48s	Baseline — single localized mode. One stationary dot at lower-center. No lattice propagation; edge dot visible.
02_dot_appears_bottom.mp4	~48s	Mode re-entry at edge. Dot reappears at bottom edge with slight positional drift. Early surface reflection visible.
03_dots_cluster_scatter.mp4	~47s	Discrete satellite cluster onset. Multiple dots in a loose, irregular cluster across the lower half. Transition from localized to discrete.
04_dots_arc_pattern.mp4	~43s	Angular preference emerging. Dots arrange into a curved arc — non-circular symmetry indicates serpentine propagation.
05_streaks_vertical.mp4	~35s	Continuous fringe to streak transition. Dots elongate into vertical streaks, suggesting partial fringe coupling.
06_streaks_full_panel.mp4	~34s	Broadest local mode extent. Multiple streaks across larger panel area — closest to collective behavior observed.

Session 2 — Dark Background / UV Suppression (videos 07–10)

File	Dur.	Description
07_discrete_cluster_dark.mp4	~41s	Discrete satellite cluster confirmed — dark field. Panel background fully dark; red dots appear as isolated points.
08_cluster_to_streaks_dark.mp4	~33s	Cluster to streak transition in dark conditions. Streak orientation non-random — same preferred axes as Session 1.
09_scattered_dots_dark.mp4	~38s	Wide-field discrete dot scatter — dark field. Dots do not blur or merge, consistent with localized mode behavior.
10_uv_suppression_red_on_blue.mp4	~36s	UV suppression trial. UV illumination shifts background to deep blue/violet. Red laser satellites remain visible.

Session 3 — Black Paper Backup (videos 11–15)

File	Dur.	Description
11_blackpaper_single_streak_onset.mp4	~40s	Single red dot with streak onset — black paper. Fringe structure suppressed; only primary emission point visible.
12_blackpaper_fringe_suppressed_dot.mp4	~34s	Fringe lines fully suppressed — black paper. Single dot with no diffuse halo. Fringe suppression occurs.
13_blackpaper_satellite_cluster_cross.mp4	~35s	Satellite cluster — cross geometry, black paper. Discrete dots in cross/plus pattern; angular preference evident.
14_blackpaper_fringe_null_dark.mp4	~31s	Near-complete fringe null state — black paper. Only faint residual dots at extreme low intensity. Fringe lines absent.
15_blackpaper_wide_discrete_scatter.mp4	~38s	Wide-field discrete scatter — black paper. Satellites spread to a broad arc as pure point-like emitters.

Session 4 — Orthogonal Polarizer Film at Input (videos 16–20)

Key finding: discrete satellite modes appear in a 1x1 array when the input polarizer is set orthogonal. Array size is not the primary driver — polarization state is.

File	Dur.	Description
16_polarizer_orthogonal_single_dot.mp4	~29s	Orthogonal polarizer baseline — single dot. Polarizer film placed orthogonal at input port of 1x1 serpentine array.
17_polarizer_orthogonal_fringe_null.mp4	~26s	Fringe fully nulled — orthogonal polarization. Only the primary dot survives. Fringe lines are polarized and suppressed.
18_polarizer_orthogonal_dot_stable.mp4	~27s	Stable isolated dot — orthogonal polarization. Primary dot holds fixed position with minimal drift. System is stable.
19_polarizer_orthogonal_discrete_onset.mp4	~27s	Discrete satellite onset — 1x1 with orthogonal polarizer. Discrete satellites appear despite 1x1 array geometry.

Technical Notes

- **Camera:** 1920x1080 @ ~30 fps (original)
- **Panel:** Diffuse white surface (LED panel / acrylic diffuser)
- **Laser:** Red (~650 nm), single-point source
- **Compressed:** 960x540, H.264 CRF 28, 15 fps — optimized for GitHub (<2 MB/file)

Analysis

Based on GPT analysis of the current footage.

The 1x1 serpentine configuration behaves more like **localized mode switching** than dynamic rotation. No global lattice propagation occurs; coupling paths are short; the edge effectively acts as the entire system. The pattern jumps between discrete states rather than flowing continuously.

Notable Finding — Angular Preference

Discrete satellite clusters are **not fully circularly symmetric**, suggesting the serpentine geometry induces an **angular preference**. The same directional satellite set reappears consistently across sessions — indicating structural involvement in **optical path selection**, not passive scattering.

Expected Changes with 6x8 Expansion

- Fringe plumes will extend and connect spatially
- Satellite activations will propagate in sequence across the array
- The current 'frozen dot' appearance is expected to diminish significantly

Summary Assessment (Sessions 1–4)

Item	Status
Continuous fringe to discrete satellite transition	Partially observed ✓
Behavior confined to local mode level	Confirmed ✓
Discrete character confirmed under dark field	Confirmed (Session 2) ✓
Fringe suppression surface-independent	Confirmed (Session 3) ✓
Red satellite survives UV background suppression	Confirmed (video 10) ✓
Discrete mode triggered by orthogonal polarizer — 1x1 sufficient	Confirmed (Session 4) ✓
Large-array collective effect	Pending — 6x8 required ✕

Conclusion: Discrete satellite mode generation is **polarization-controlled**, not array-size-dependent. Orthogonal polarizer input is a sufficient condition for discrete behavior even in a 1x1 configuration. 6x8 expansion remains necessary for large-scale collective propagation effects.